

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 6

For the tutorial in Week 7 on 29 February, let us discuss

- Ex. 8.2.11, 8.2.22, 8.2.24, 8.2.25, 8.2.26, additional exercises.

Ex. 8.2.11. Find the solution to $a_n = 2a_{n-1} + 5a_{n-2} - 6a_{n-3}$ with $a_0 = 7, a_1 = -4, a_2 = 8$.

Solution: This is a linear homogenous recurrent with degree 3. The characteristic equation is $r^3 - 2r^2 - 5r + 6 = 0$. Factorizing the expression on the left side, we see that

$$(r - 1)(r + 2)(r - 3) = 0.$$

Thus the characteristic equation has 3 distinct roots $r_1 = 1, r_2 = -2, r_3 = 3$. Then $a_n = h_1 \cdot 1^n + h_2 \cdot (-2)^n + h_3 \cdot 3^n$. The initial conditions are:

$$\begin{aligned} 7 &= a_0 = h_1 + h_2 + h_3, \\ -4 &= a_1 = h_1 - 2h_2 + 3h_3, \\ 8 &= a_2 = h_1 + 4h_2 + 9h_3, \end{aligned}$$

solving which leads to $h_1 = 5, h_2 = 3, h_3 = -1$. Hence, for $n \geq 0$,

$$a_n = 5 + 3 \cdot (-2)^n - 3^n.$$

Ex. 8.2.22.

- (a) Find all solutions of the recurrence relation $a_n = 2a_{n-1} + 2n^2$ for $n \geq 2$.
- (b) Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 4$.

Solution:

- (a) The associated homogeneous recurrence is $b_n = 2b_{n-1} = \dots = 2^n b_0 = h \cdot 2^n$.

Since 1 is NOT a solution of the characteristic equation, let $a_n^{(p)} = cn^2 + dn + e$ be a particular solution of the recurrence. Then

$$cn^2 + dn + e = 2(c(n-1)^2 + d(n-1) + e) + 2n^2.$$

Expanding and collecting terms, we have

$$(c + 2)n^2 + (d - 4c)n + (2c - 2d + e) = 0,$$

whence $c = -2, d = -8$ and $e = -12$.

Hence $a_n^{(p)} = -2n^2 - 8n - 12$ is a particular solution, and all solutions of the recurrence relation are of the form

$$a_n = h \cdot 2^n - 2n^2 - 8n - 12, \quad \text{for } n \geq 2.$$

- (b) By initial condition, $a_1 = 4$, substituting it into the solution above, we obtain

$$a_1 = 4 = h \cdot 2^1 - 2 \cdot 1^2 - 8 \cdot 1 - 12 = 2h - 22.$$

Hence we have $h = 13$. Hence, $a_n = 13 \cdot 2^n - 2n^2 - 8n - 12$ for $n \geq 1$.

Ex. 8.2.24.

- (a) Find all solutions of the recurrence relation $a_n = -5a_{n-1} - 6a_{n-2} + 42 \cdot 4^n$.
 (b) Find the solution of the recurrence relation in part (a) with initial conditions $a_1 = 56$ and $a_2 = 278$.

Solution:

- (a) The associated homogenous recurrence is

$$b_n = -5b_{n-1} - 6b_{n-2},$$

which is of degree 2. The characteristic equation is $r^2 + 5r + 6 = (r+2)(r+3) = 0$, which has 2 distinct roots $r_1 = -2$ and $r_2 = -3$. Then

$$b_n = h_1 \cdot (-2)^n + h_2 \cdot (-3)^n.$$

On the other hand, $F(n) = 42 \cdot 4^n$, which is an exponential, hence there is a particular solution taking exponential form. Since 4 is NOT a root of the characteristic equation, we let

$$a_n^{(p)} = c \cdot 4^n.$$

Substituting this back to the original equation gives

$$c \cdot 4^n = -5 \cdot c \cdot 4^{n-1} - 6 \cdot c \cdot 4^{n-2} + 42 \cdot 4^n.$$

Dividing throughout by 4^{n-2} , we arrive at

$$16c = -20c - 6c + 42 \times 16.$$

Hence $c = 16$, and so $a_n^{(p)} = 4^{n+2}$. Hence, all solutions of the recurrence relation are of the form

$$a_n = b_n + a_n^{(p)} = h_1 \cdot (-2)^n + h_2 \cdot (-3)^n + 4^{n+2}.$$

- (b) The initial conditions give

$$56 = a_1 = -2h_1 - 3h_2 + 64,$$

$$278 = a_2 = 4h_1 + 9h_2 + 256,$$

solving which leads to $h_1 = 1$ and $h_2 = 2$. Therefore,

$$a_n = (-2)^n + 2 \cdot (-3)^n + 4^{n+2}, \quad \text{for } n \geq 1.$$

Ex. 8.2.25. Find the solution of the recurrence relation $a_n = 4a_{n-1} - 3a_{n-2} + 2^n + n + 3$ with $a_0 = 1$ and $a_1 = 4$.

(Hint: Find $a_n^{(p1)}$ which takes care of the exponential part, i.e., a particular solution of $4a_{n-1} - 3a_{n-2} + 2^n$. Next, find $a_n^{(p2)}$ which takes care of the polynomial part, i.e., a particular solution of $4a_{n-1} - 3a_{n-2} + n + 3$. Finally, add them up, $a_n^{(p)} = a_n^{(p1)} + a_n^{(p2)}$.)

Solution:

The associated homogenous recurrence is

$$b_n = 4b_{n-1} - 3b_{n-2},$$

which is of degree 2. The characteristic equation is $r^2 - 4r + 3 = (r-1)(r-3) = 0$, which has 2 distinct roots $r_1 = 1$ and $r_2 = 3$. Then

$$b_n = h_1 \cdot 1^n + h_2 \cdot 3^n = h_1 + h_2 \cdot 3^n.$$

On the other hand, $F(n) = 2^n + n + 3$, which is an exponential plus a polynomial, hence the particular solution will also be an exponential plus a polynomial. Let's solve it one by one, for the exponential since 2 is NOT a solution to the characteristic equation

$$a_n^{(p1)} = c \cdot 2^n.$$

Substituting this back to the equation $a_n = 4a_{n-1} - 3a_{n-2} + 2^n$ leads to

$$c \cdot 2^n = 4 \cdot c \cdot 2^{n-1} - 3 \cdot c \cdot 2^{n-2} + 2^n,$$

which gives $c = -4$ and $a_n^{(p1)} = -2^{n+2}$.

For the polynomial part of $F(n)$, since 1 IS a solution to the characteristic equation

$$a_n^{(p2)} = n \cdot (d \cdot n + e) = dn^2 + en.$$

Substituting this back to the equation $a_n = 4a_{n-1} - 3a_{n-2} + n + 3$ leads to

$$dn^2 + en = 4[d(n-1)^2 + e(n-1)] - 3[d(n-2)^2 + e(n-2)] + n + 3,$$

which is equivalent with

$$(4d + 1) \cdot n + (-8d + 2e + 3) = 0.$$

This implies that

$$4d + 1 = 0 \quad \text{and} \quad -8d + 2e + 3 = 0.$$

Hence $d = -1/4$ and $e = -5/2$, and so $a_n^{(p2)} = -n^2/4 - 5n/2$. Hence

$$a_n = b_n + a_n^{(p1)} + a_n^{(p2)} = h_1 + h_2 \cdot 3^n - 2^{n+2} - n^2/4 - 5n/2.$$

The initial conditions imply

$$1 = a_0 = h_1 + h_2 - 4,$$

$$4 = a_1 = h_1 + 3h_2 - 8 - 1/4 - 5/2,$$

solving which leads to $h_1 = 1/8$ and $h_2 = 39/8$, and

$$a_n = b_n + a_n^{(p1)} + a_n^{(p2)} = 1/8 + 39/8 \cdot 3^n - 2^{n+2} - n^2/4 - 5n/2, \quad \text{for } n \geq 0.$$

Ex. 8.2.26. Let a_n be the sum of the first n perfect squares, that is $a_n = \sum_{k=1}^n k^2$. Using linear non-homogeneous recurrence relations to determine a formula for a_n .

(Hint: observe the relation between a_n and a_{n-1} , i.e. $a_n = a_{n-1} + ?$)

Solution:

Obviously the n th term of the sequence comes from the $(n-1)$ th term by adding n^2 . That is, we have

$$a_n = a_{n-1} + n^2.$$

Also clearly $a_1 = 1$.

To solve this recurrence relation, we easily see that the homogeneous solution is $a_n = \alpha$ for some constant α .

Since the non-homogeneous term is a second degree polynomial, and 1 is a characteristic root of the associated homogeneous relation, we need a particular solution of the form $a_n = cn^3 + dn^2 + en$. Plugging this into the recurrence relation gives

$$cn^3 + dn^2 + en = c(n-1)^3 + d(n-1)^2 + e(n-1) + n^2.$$

Expanding and collecting terms, we have

$$(3c - 1)n^2 + (-3c + 2d)n + (c - d + e) = 0$$

and so

$$3c - 1 = 0, \quad -3c + 2d = 0, \quad \text{and} \quad c - d + 3 = 0.$$

Hence $c = 1/3$, $d = 1/2$ and $e = 1/6$. Thus

$$a_n^{(p)} = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.$$

So the general solution is

$$a_n = \alpha + \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.$$

Next, by applying the initial condition $a_1 = 1$, we see that $\alpha = 0$. Finally, expressing the terms of a_n using a common denominator, and write our solution in the familiar form

$$a_n = \frac{n(n+1)(2n+1)}{6}, \quad \text{for } n \geq 1.$$

Additional Exercise 1. Solve the recurrence relation $a_n = a_{n-1}^2/a_{n-2}$ if $a_0 = 1$ and $a_1 = 2$. (*Hint:* take logarithms of both sides to obtain a recurrence relation for the sequence $\{\log a_n\}$. Alternatively, one can make an observation on the ratio a_n/a_{n-1} by working out the values a_1/a_0 , a_2/a_1 , a_3/a_2 , etc.)

Solution: Following the hint, we let $b_n = \log_2 a_n$. Then the recurrence relation becomes $b_n = 2b_{n-1} - b_{n-2}$, with initial conditions $b_0 = 0$, $b_1 = 1$.

Solve this in usual manner. The characteristic equation is $r^2 - 2r + 1 = 0$ with the same root $r_0 = 1$ with multiplicity 2. Thus b_n has general form

$$b_n = (h_1 + h_2n) \cdot 1^n = h_1 + h_2n.$$

Plugging in the initial conditions gives us $h_1 = 0$ and $h_2 = 1$. Hence we have $b_n = n$. Therefore, the solution to the original problem is then $a_n = 2^n$, for $n \geq 0$.

Additional Exercise 2. Show that the Fibonacci numbers satisfy the recurrence relation $f_n = 5f_{n-4} + 3f_{n-5}$ for $n = 5, 6, 7, \dots$ together with the initial conditions $f_0 = 0$, $f_1 = 1$, $f_2 = 1$, $f_3 = 2$, $f_4 = 3$.

Solution:

The Fibonacci sequence fulfills $f_n = f_{n-1} + f_{n-2}$, hence $f_{n-1} = f_{n-2} + f_{n-3}$ (by replacing n with $n-1$). Substitute the latter into the former, we obtain

$$\begin{aligned} f_n &= (f_{n-2} + f_{n-3}) + f_{n-2} \\ &= 2f_{n-2} + f_{n-3}. \end{aligned}$$

Similarly $f_{n-2} = f_{n-3} + f_{n-4}$. Hence

$$\begin{aligned} f_n &= 2(f_{n-3} + f_{n-4}) + f_{n-3} \\ &= 3f_{n-3} + 2f_{n-4} \\ &= 3(f_{n-4} + f_{n-5}) + 2f_{n-4} \\ &= 5f_{n-4} + 3f_{n-5}. \end{aligned}$$

Additional exercise 3. Suppose that the sequences $(s_0, s_1, s_2, \dots)$ and $(t_0, t_1, t_2, \dots)$ both satisfy the same linear homogeneous recurrence relation with constant coefficients: for all integers $k \geq 2$, $s_k = 5s_{k-1} - 4s_{k-2}$ and $t_k = 5t_{k-1} - 4t_{k-2}$. Define a new sequence $(u_0, u_1, u_2, \dots)$ where $u_i = 2s_i + 3t_i$. Show that $(u_0, u_1, u_2, \dots)$ also satisfies the same relation (i.e. $u_k = 5u_{k-1} - 4u_{k-2}$).

Solution:

Multiplying 2 to both sides of $s_k = 5s_{k-1} - 4s_{k-2}$ gives

$$2s_k = 10s_{k-1} - 8s_{k-2}.$$

Multiplying 3 to both sides of $t_k = 5t_{k-1} - 4t_{k-2}$ gives

$$3t_k = 15t_{k-1} - 12t_{k-2}.$$

Adding the two equations gives

$$\begin{aligned} u_k &= 2s_k + 3t_k \\ &= 10s_{k-1} - 8s_{k-2} + 15t_{k-1} - 12t_{k-2} \\ &= 5(2s_{k-1} + 3t_{k-1}) - 4(2s_{k-2} + 3t_{k-2}) = 5u_{k-1} - 4u_{k-2}. \end{aligned}$$

Hence u_k also satisfies the same relation.

The following are optional exercises. We will not be discussing them during tutorial.

Ex. 8.2.14. Solve the recurrence relation $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ with $a_0 = -5, a_1 = 4, a_2 = 88$.

Solution: This is a linear homogenous recurrent with degree 3. The characteristic equation is $r^3 - 6r^2 + 12r - 8 = 0$. Factorizing the left side gives

$$(r - 2)^3 = 0,$$

which shows that the characteristic equation has 1 root, $r_0 = 2$ with multiplicity 3. Then $a_n = (h_1 \cdot n^2 + h_2 \cdot n + h_3) \cdot 2^n$. The initial conditions are:

$$\begin{aligned} -5 &= a_0 = h_3, \\ 4 &= a_1 = (h_1 + h_2 + h_3) \cdot 2, \\ 88 &= a_2 = (4h_1 + 2h_2 + h_3) \cdot 4, \end{aligned}$$

solving which leads to $h_1 = 13/2, h_2 = 1/2, h_3 = -5$. Hence, for $n \geq 0$,

$$a_n = (13n^2 + n - 10) \cdot 2^{n-1}.$$

Ex. 8.2.15. Solve the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with $a_0 = 5, a_1 = -9, a_2 = 15$.

Solution: This is a linear homogenous recurrent with degree 3. The characteristic equation is $r^3 + 3r^2 + 3r + 1 = 0$. Factorizing the left side gives

$$(r + 1)^3 = 0,$$

which shows that the characteristic equation has 1 distinct root, $r_0 = -1$ with multiplicity 3. Then $a_n = (h_1 \cdot n^2 + h_2 \cdot n + h_3) \cdot (-1)^n$. The initial conditions are:

$$\begin{aligned} 5 &= a_0 = h_3, \\ -9 &= a_1 = (h_1 + h_2 + h_3) \cdot (-1), \\ 15 &= a_2 = (4h_1 + 2h_2 + h_3), \end{aligned}$$

solving which leads to $h_1 = 1, h_2 = 3, h_3 = 5$. Hence, for $n \geq 1$,

$$a_n = (n^2 + 3n + 5) \cdot (-1)^n.$$

Ex. 8.2.17 and 8.2.18. What is the general form of the solutions of a linear homogeneous recurrence relation if its characteristic equation has roots

- (a) $1, 1, 1, 1, -2, -2, -2, 3, 3, -4$?
 (b) $-1, -1, -1, 2, 2, 5, 5, 7$?

Solution:

- (a) There are 4 distinct roots with multiplicity $(r_1 = 1, m_1 = 4), (r_2 = -2, m_2 = 3), (r_3 = 3, m_3 = 2), (r_4 = -4, m_4 = 1)$. Then the solution is of the form with constants h

$$\begin{aligned} a_n &= (h_{1,0} \cdot n^3 + h_{1,1} \cdot n^2 + h_{1,2} \cdot n + h_{1,3}) \cdot 1^n + \\ &\quad (h_{2,0} \cdot n^2 + h_{2,1} \cdot n + h_{2,2}) \cdot (-2)^n + \\ &\quad (h_{3,0} \cdot n + h_{3,1}) \cdot 3^n + \\ &\quad (h_{4,0}) \cdot (-4)^n. \end{aligned}$$

- (b) There are 4 distinct roots with multiplicity $(r_1 = -1, m_1 = 3), (r_2 = 2, m_2 = 2), (r_3 = 5, m_3 = 2), (r_4 = 7, m_4 = 1)$. Then the solution is of the form with constants h

$$\begin{aligned} a_n &= (h_{1,0} \cdot n^2 + h_{1,1} \cdot n + h_{1,2}) \cdot (-1)^n + \\ &\quad (h_{2,0} \cdot n + h_{2,1}) \cdot 2^n + \\ &\quad (h_{3,0} \cdot n + h_{3,1}) \cdot 5^n + \\ &\quad (h_{4,0}) \cdot 7^n. \end{aligned}$$

Ex. 8.2.23.

- (a) Find all solutions of the recurrence relation $a_n = 2a_{n-1} + 3^n$ for $n \geq 2$.
 (b) Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 5$.

Solution:

- (a) The associated homogeneous recurrence is $b_n = 2b_{n-1} = \dots = 2^n b_0 = h \cdot 2^n$. Since 3 is NOT a solution to the characteristic equation, let $a_n^{(p)} = c3^n$ is a particular solution of the recurrence. Then

$$c3^n = 2c3^{n-1} + 3^n.$$

Dividing throughout by 3^{n-1} , we arrive at

$$3c = 2c + 3,$$

which implies $c = 3$. Hence, all solutions of the recurrence relation are of the form

$$a_n = h \cdot 2^n + 3 \cdot 3^n.$$

(b) By initial condition, $a_1 = 5$, substituting it into the solution above, we obtain

$$a_1 = 5 = h \cdot 2^1 + 3 \cdot 3^1 = 2h + 9.$$

Hence $h = -2$. Hence, $a_n = -2^{n+1} + 3^{n+1}$ for $n \geq 1$.

Answer.

Ex. 8.2.11. $a_n = 5 + 3 \cdot (-2)^n - 3^n$ for $n \geq 0$.

Ex. 8.2.22. $a_n = h \cdot 2^n - 2n^2 - 8n - 12$ for $n \geq 2$; $a_n = 13 \cdot 2^n - 2n^2 - 8n - 12$ for $n \geq 1$.

Ex. 8.2.24. $a_n = h_1 \cdot (-2)^n + h_2 \cdot (-3)^n + 4^{n+2}$; $a_n = (-2)^n + 2 \cdot (-3)^n + 4^{n+2}$ for $n \geq 1$.

Ex. 8.2.25. $a_n = 1/8 + 39/8 \cdot 3^n - 2^{n+2} - n^2/4 - 5n/2$ for $n \geq 0$.

Ex. 8.2.26. $a_n = n(n+1)(2n+1)/6$ for $n \geq 1$.

Additional Exercise 1. $a_n = 2^n$ for $n \geq 0$.